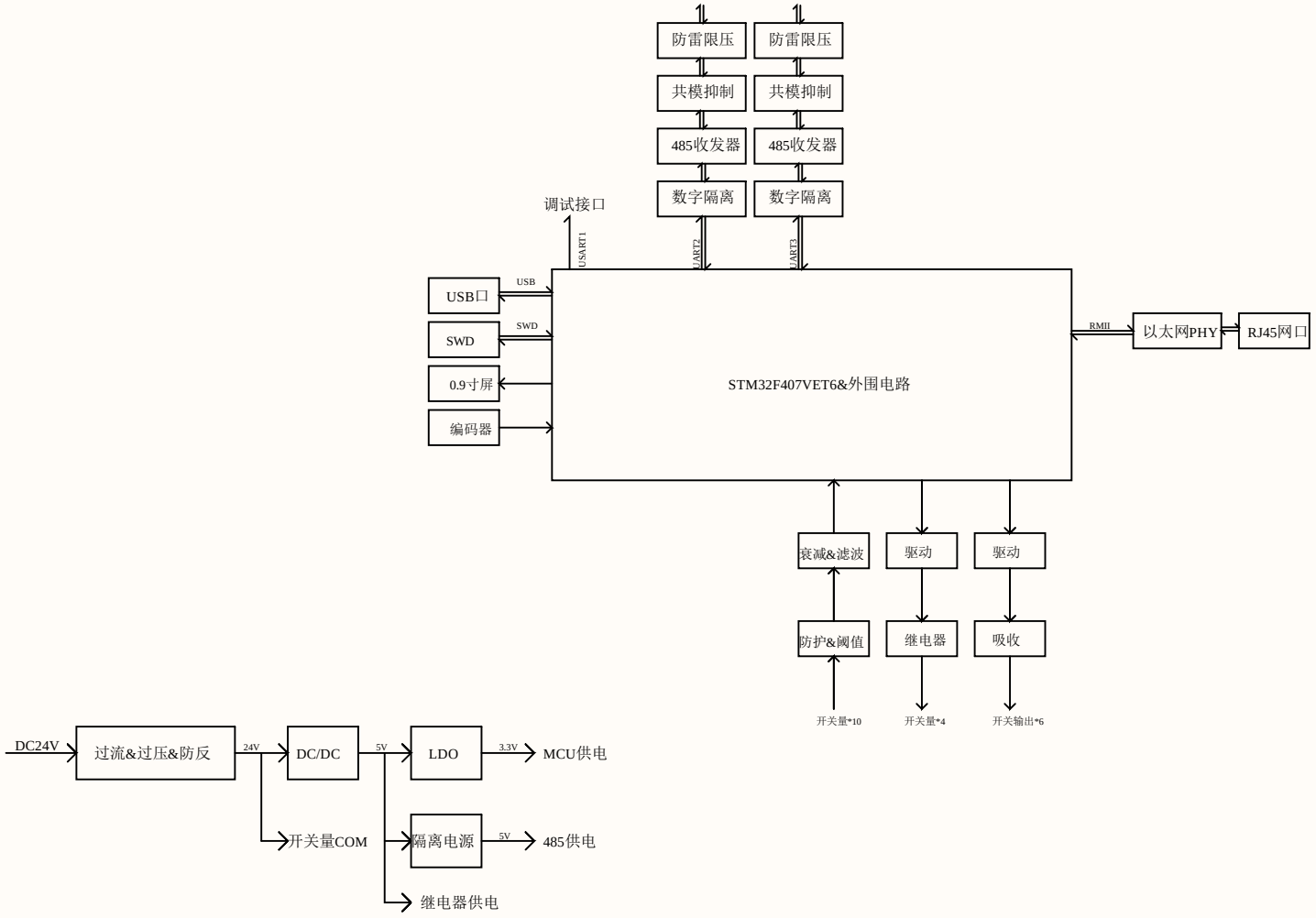
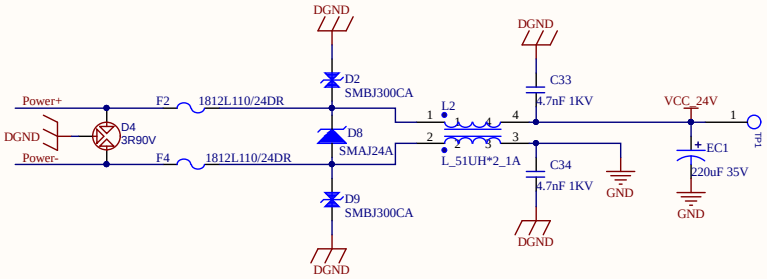
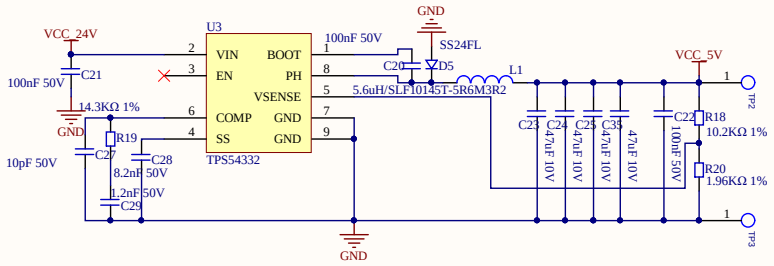
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V1.0	2022.09.20	桑牧丹	新建

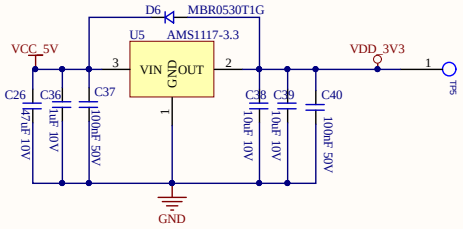
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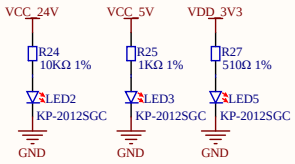


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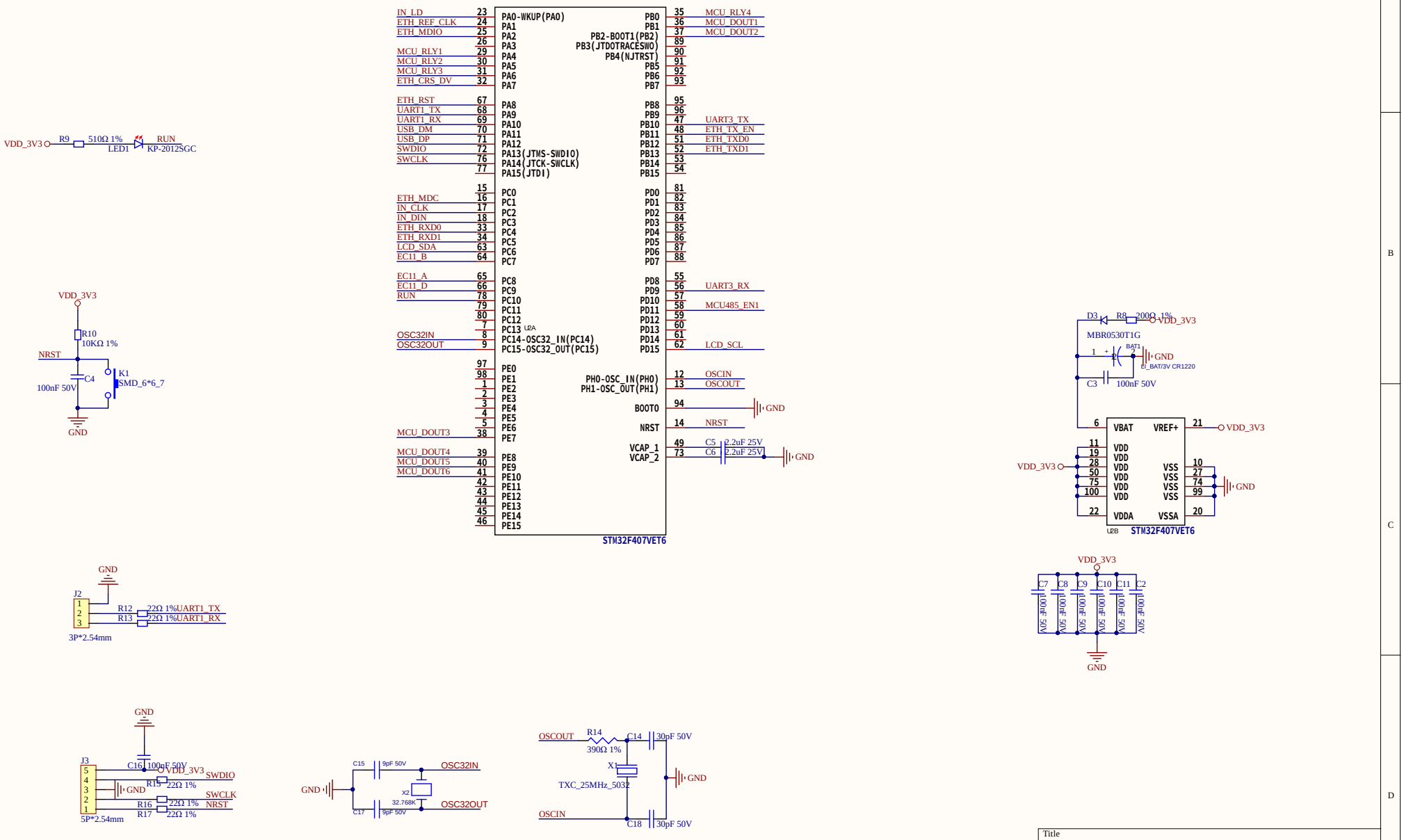




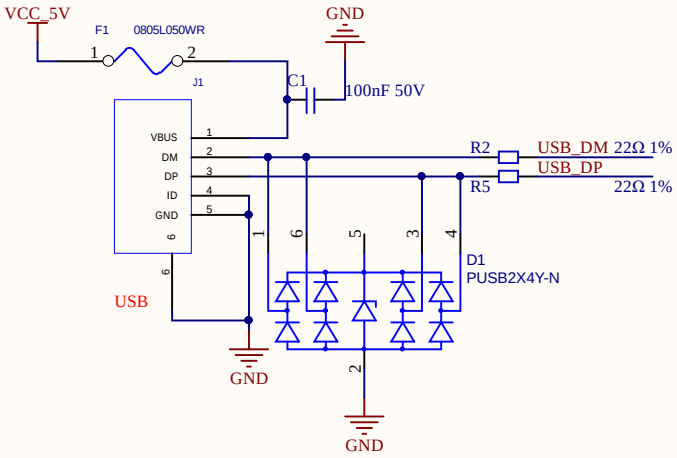




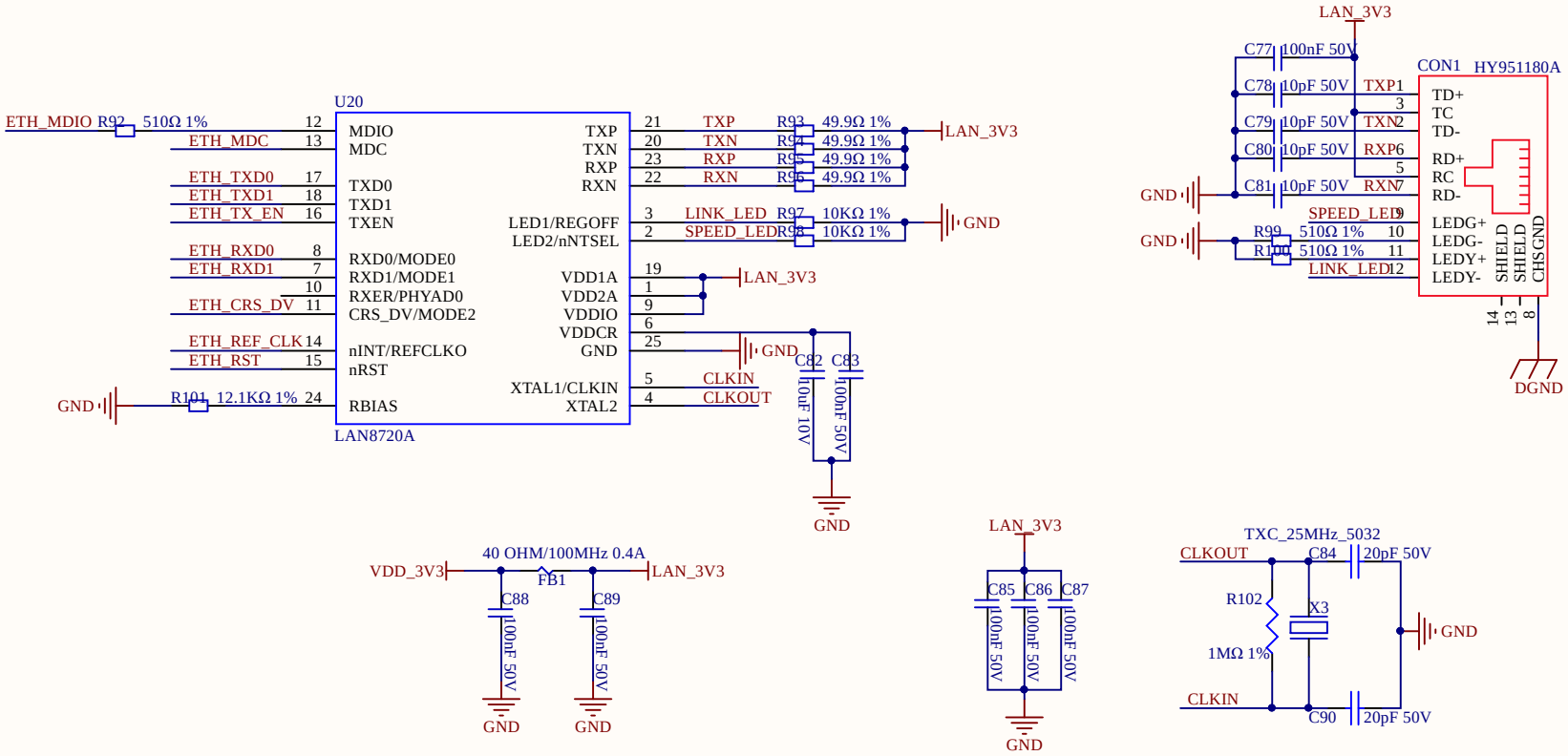
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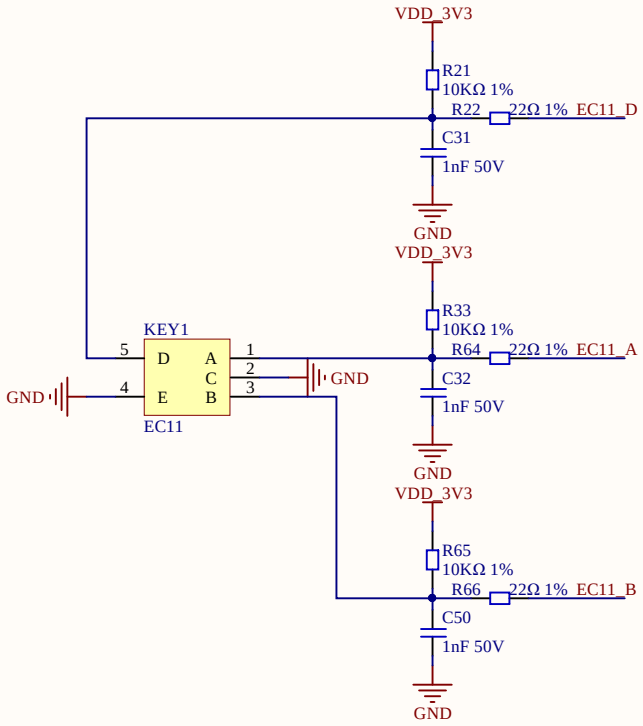
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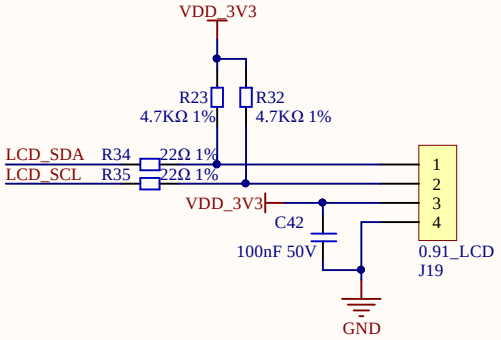


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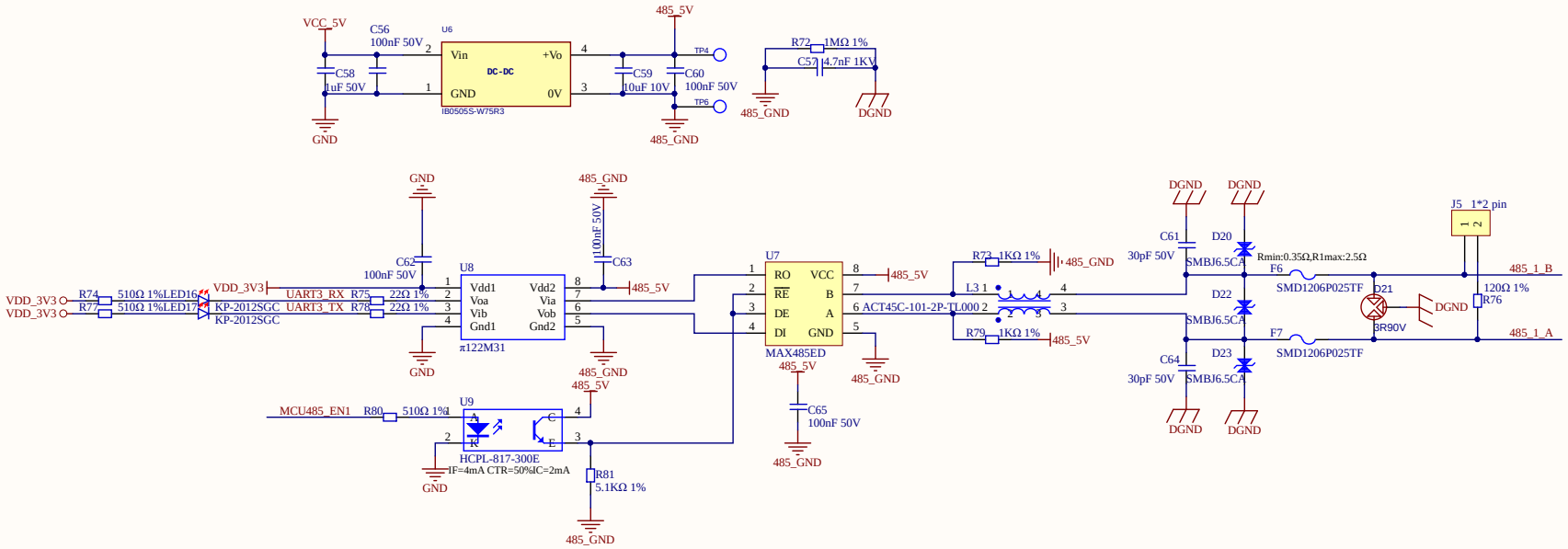


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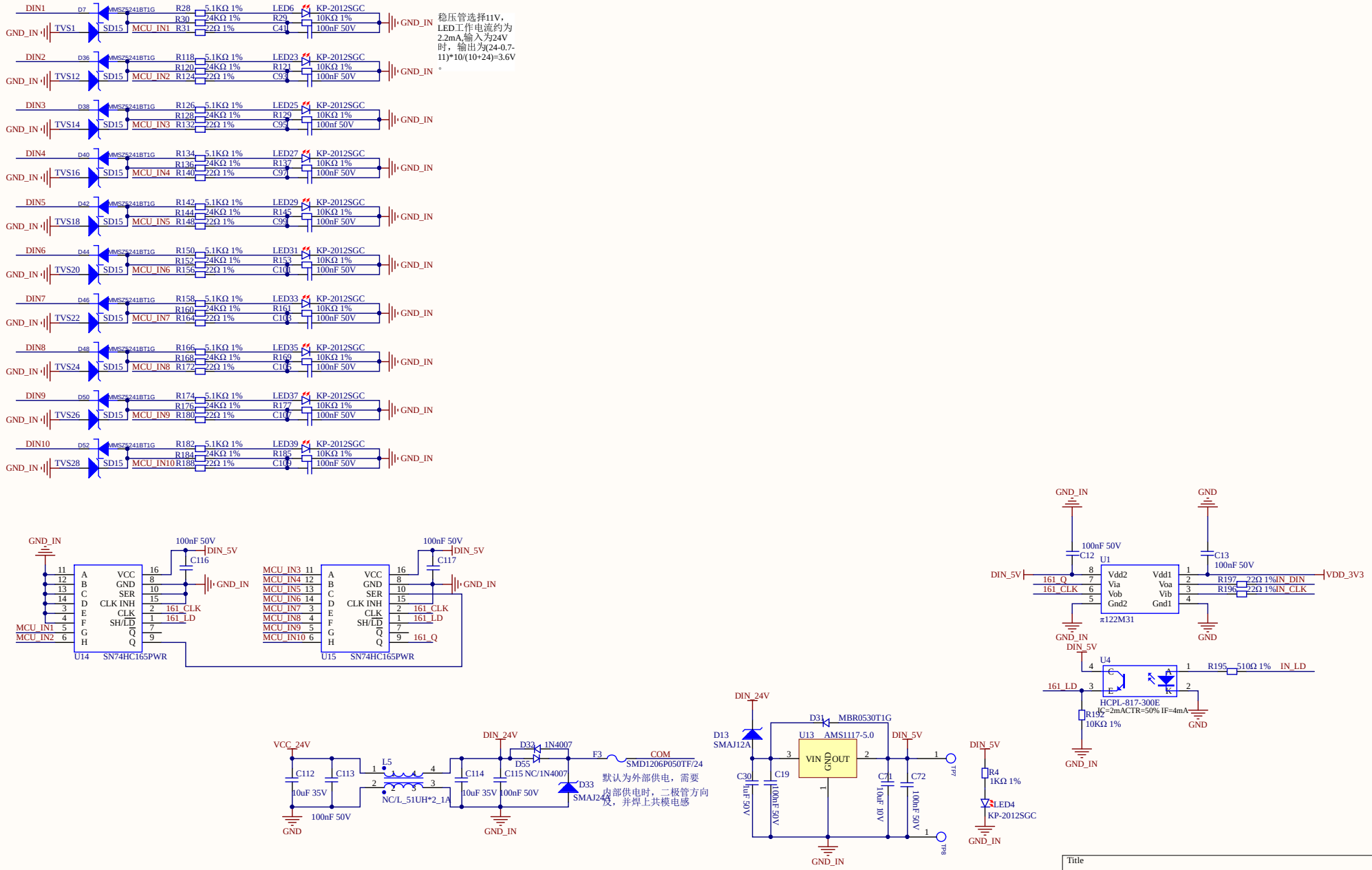


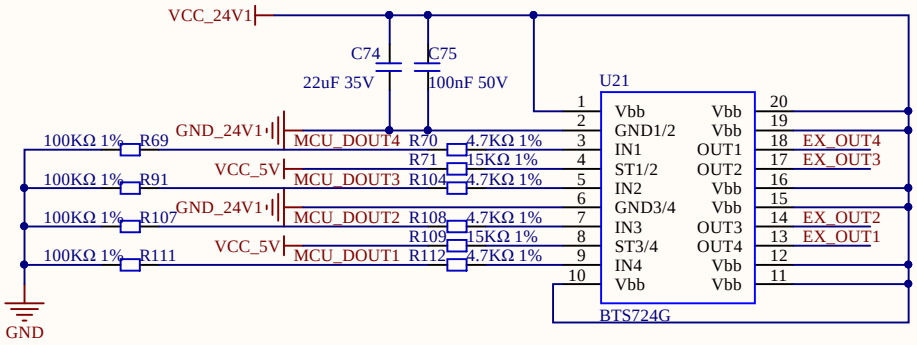


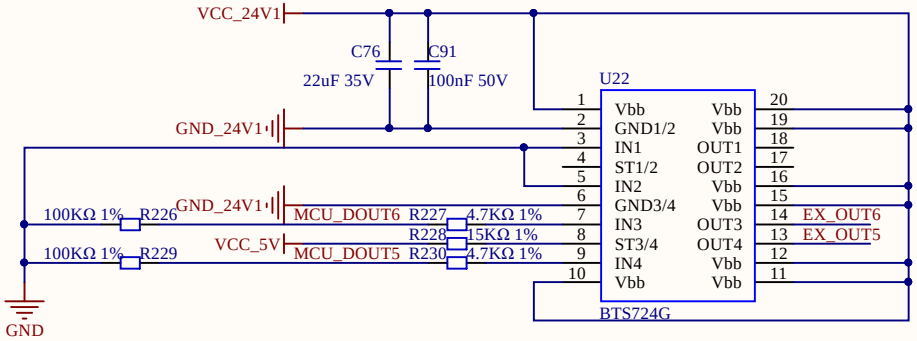
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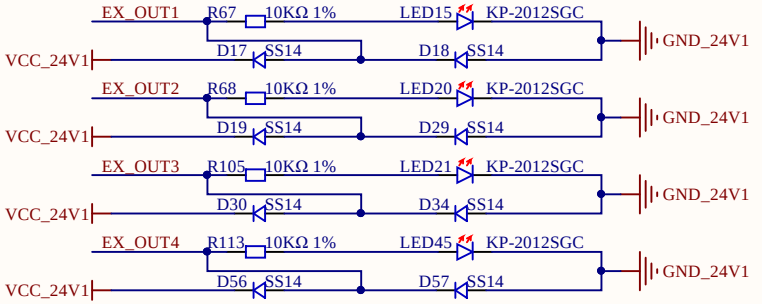


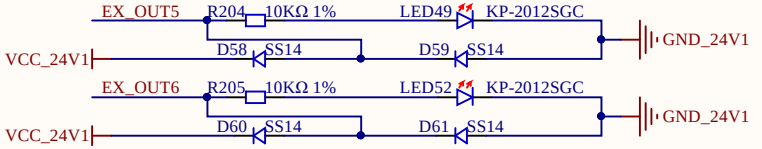
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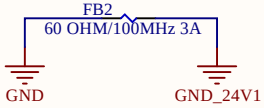

DIN1 D7 SMMSZ5241BT1G R28 5.1KΩ 1% LED6 KP-2012SGC
R30 24KΩ 1% R29 10KΩ 1%
GND_IN TVS1 SD15 MCU_IN1 R31 22Ω 1% C41 100nF 50V GND_IN
DIN2 D36 R118 5.1KΩ 1% LED23 R121 10KΩ 1% C93 100nF 50V
DIN3 D38 R126 5.1KΩ 1% LED25 R129 10KΩ 1% C95 100nf 50V
DIN4 D40 R134 5.1KΩ 1% LED27 R137 10KΩ 1% C97 100nF 50V
DIN5 D42 R142 5.1KΩ 1% LED29 R145 10KΩ 1% C99 100nF 50V
DIN6 D44 R150 5.1KΩ 1% LED31 R153 10KΩ 1% C101 100nF 50V
DIN7 D46 R158 5.1KΩ 1% LED33 R161 10KΩ 1% C103 100nF 50V
DIN8 D48 R166 5.1KΩ 1% LED35 R169 10KΩ 1% C105 100nF 50V
DIN9 D50 R174 5.1KΩ 1% LED37 R177 10KΩ 1% C107 100nF 50V
DIN10 D52 R182 5.1KΩ 1% LED39 R185 10KΩ 1% C109 100nF 50V
R184 24KΩ 1%
TVS18 TVS20 TVS22 TVS24 TVS26 TVS28
MCU_IN2 MCU_IN3 MCU_IN4 MCU_IN5 MCU_IN6 MCU_IN7 MCU_IN8 MCU_IN9 MCU_IN10
稳压管选择11V，LED工作电流约为2.2mA,输入为24V时，输出为(24-0.7-11)*10/(10+24)=3.6V。
GND_IN 100nF 50V C116 DIN_5V GND_IN
U14 SN74HC165PWR
C117
U15 SN74HC165PWR
161_CLK 161_LD 161_Q
VCC_24V C112 C113 10uF 35V 100nF 50V GND
L5 NC/L_51UH*2_1A
C114 10uF 35V C115 NC/1N4007 100nF 50V GND_IN
D32 1N4007 D55 D33 SMAJ24A
F3 SMD1206P050TF/24 COM
默认为外部供电，需要内部供电时，二极管方向反，并焊上共模电感
DIN_24V D13 SMAJ12A D31 MBR0530T1G
U13 AMS1117-5.0 VIN GND OUT
C30 10uF 50V C19 100nF 50V C71 10uF 10V C72 100nF 50V
TP7 TP8
R4 1KΩ 1% LED4 KP-2012SGC
C12 100nF 50V C13 100nF 50V
U1 π122M31 Vdd2 Via Vob Gnd2 Vdd1 Voa Vib Gnd1
R197 22Ω 1% IN_DIN R196 22Ω 1% IN_CLK VDD_3V3
U4 HCPL-817-300E IC=2mACTR=50% IF=4mA
R195 510Ω 1% IN_LD R194 10KΩ 1%
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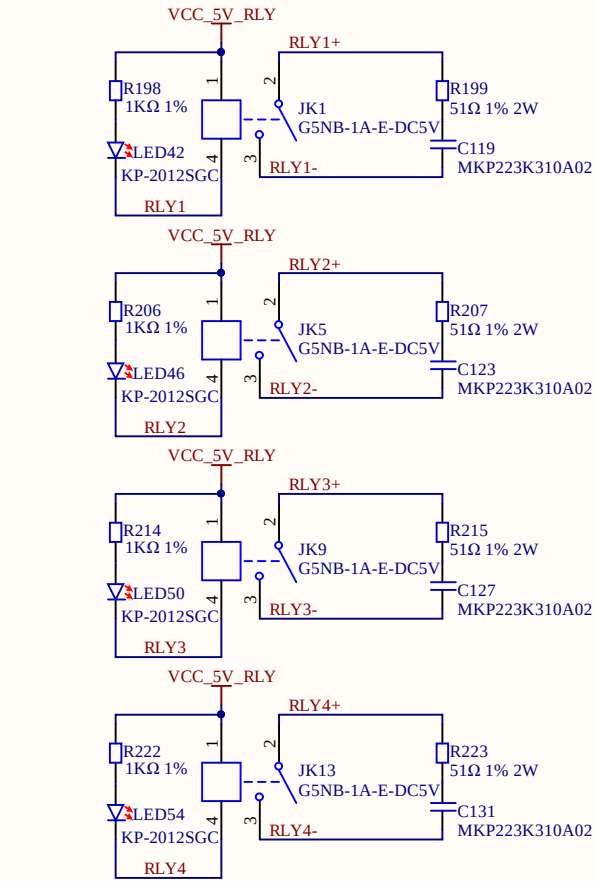


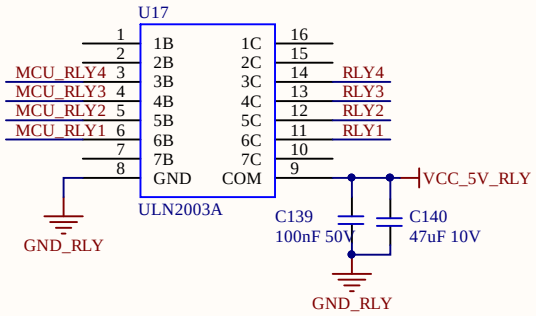


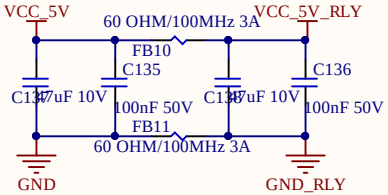




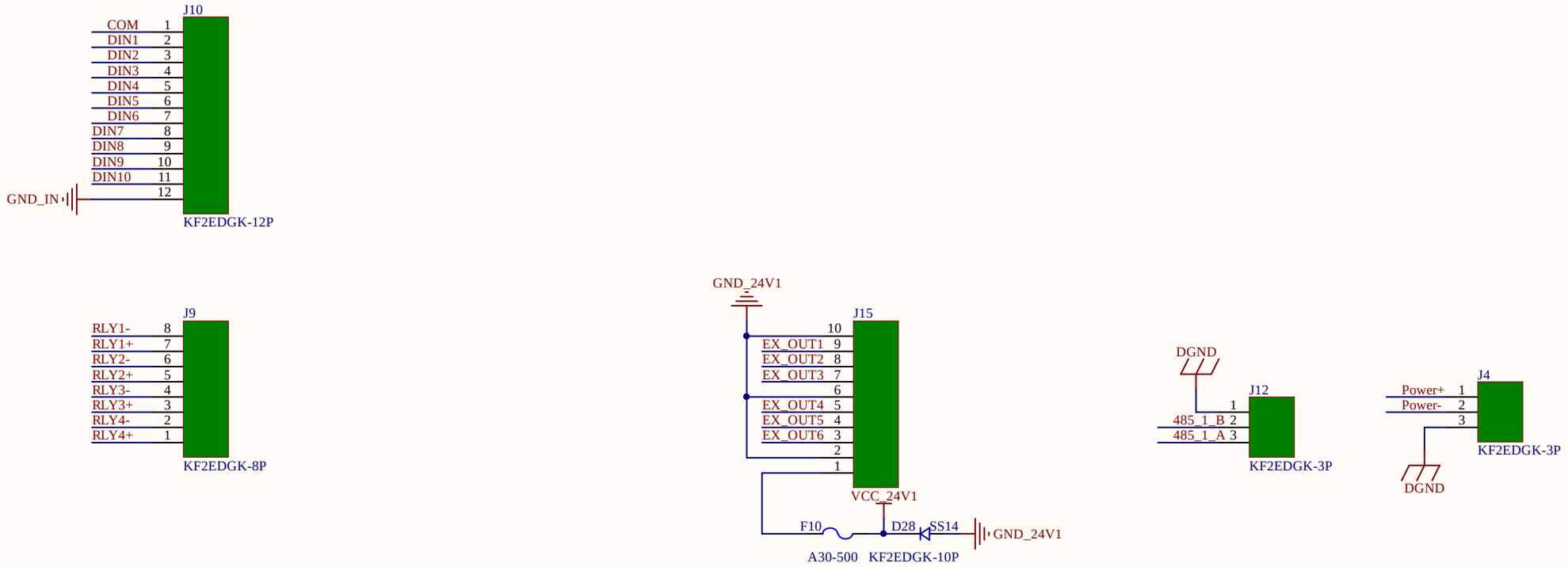
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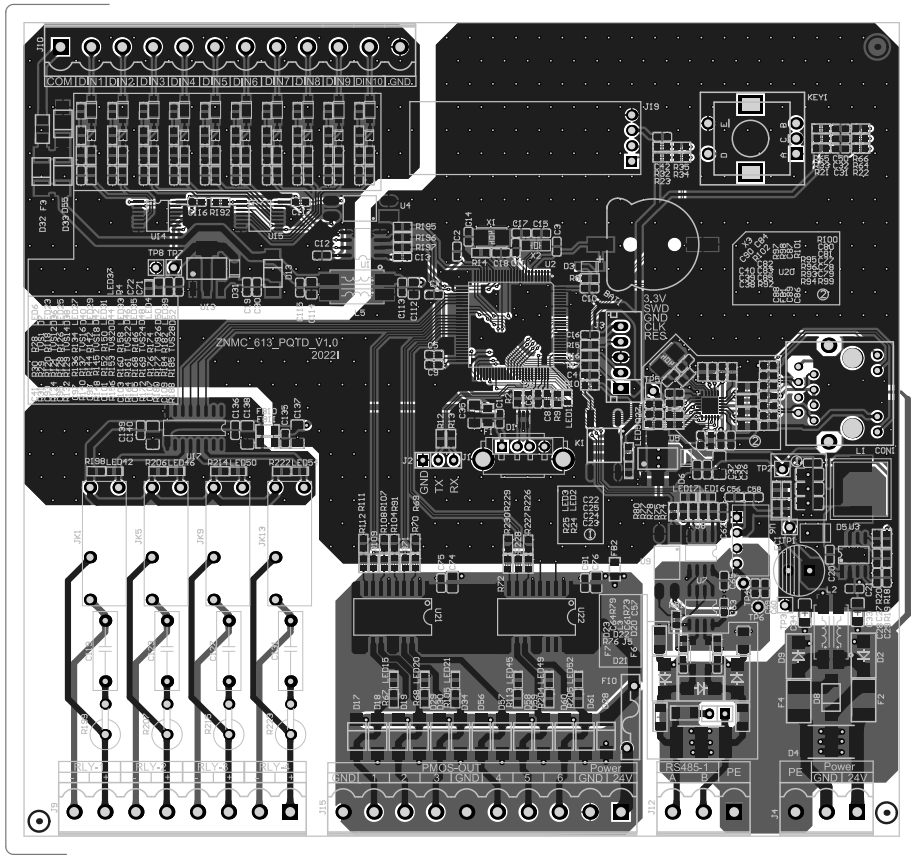




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Size A4	Number	Revision
Date: 2022/9/20		Sheet of
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COM DIN1 DIN2 DIN3 DIN4 DIN5 DIN6 DIN7 DIN8 DIN9 DIN10 GND
ZNMC_613_PQTD_V1.0
2022
3.3V
SWD
GND
CLK
RES
TX
RX
GND
RLY-1
RLY-2
RLY-3
RLY-4
PMOS-OUT
GND
Power
GND 24V
RS485-1
PE
PE
Power
GND 24V